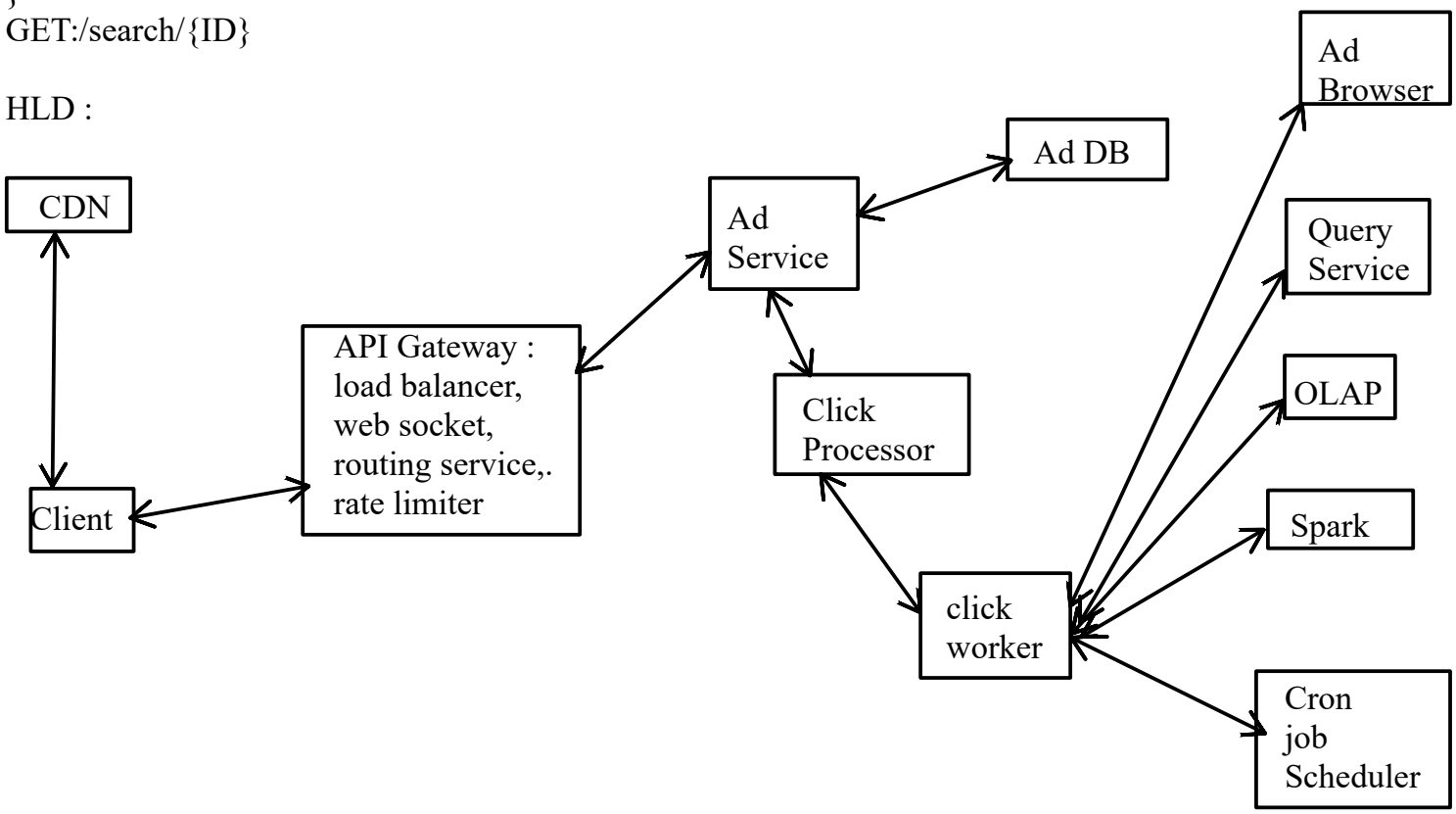
Ad Click Aggregator :

Functional requirements:
user should be able to click on a particular Ad & redirect into particular website accordingly ,
Ad should be customised based on the user's activity,

Non-Functional requirements:
Data-partitioning,
load balancing,
fault tolerance,
vertical scalability,
horizontal scalability,
low latency,
high throughput,
CAP theorem(Here eventual consistency is fine , but here we need 99.99% uptime , i.e Availability>>Consistency)

Core Entities:
user ,
Ad,
Activity,
history

API :
POST:/click/Ad
{
userID,
URL
}
GET:/search/{ID}

HLD :